

Ayurveda IPR Assistant — Complete Data Specification Guide

What to Scrape, How to Scrape, Exact Data Fields, Unified Schema for RAG

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1. UNIFIED DATA FORMAT — The Master Schema

IMPORTANT

Every piece of scraped data, regardless of source, MUST be converted into this unified JSON format before entering the RAG pipeline. This ensures consistent chunking, embedding, retrieval, and citation.

1.1 The Universal Document Object (UDO)

Every scraped item becomes one or more UDO records. This is the ONLY format your RAG system reads.

```

{
  "doc_id": "IND-PAT-1970-S003P",
  "doc_type": "statute_section",
  "title": "Section 3(p) - Inventions relating to traditional knowledge",
  "content": "The full text of this section goes here. An invention which in effect, is traditional knowledge or which is an aggregation or duplication of known properties of traditionally known component or components.",
  "content_plain": "Same as above but stripped of all formatting",

  "source": {
    "name": "Patents Act, 1970",
    "url": "https://indiacode.nic.in/handle/123456789/1392",
    "access_date": "2026-08-15",
    "version": "2024-amendment",
    "page_number": null,
    "volume": null
  },

  "hierarchy": {
    "level_1": "Patents Act, 1970",
    "level_2": "Chapter II - Inventions Not Patentable",
    "level_3": "Section 3",
    "level_4": "Clause (p)",
    "level_5": null
  },

  "metadata": {
    "jurisdiction": "India",
    "language": "en",
    "category": "statute",
    "sub_category": "patent_law",
    "effective_date": "2005-01-01",
    "last_amended": "2024-03-15",
    "is_current": true,
    "tags": ["patent_bar", "traditional_knowledge", "section_3p", "AYUSH"],
    "cross_references": [
      "IND-PAT-1970-S003D",
      "IND-PAT-1970-S003E",
      "IND-TKDL-POINTER"
    ],
    "related_statutes": ["Biological Diversity Act, 2002", "D&C Act, 1940"]
  },

  "rag_config": {
    "chunk_strategy": "section_boundary",
    "max_tokens": 1500,
    "overlap_tokens": 200,
    "embedding_model": "BGE-M3",
    "importance_score": 0.95,
    "citation_format": "[Section 3(p), Patents Act, 1970]"
  },

  "checksum": "sha256:a1b2c3d4e5f6..."
}

```

1.2 Document Types (doc_type values)

doc_type	What It Represents	Source Category
statute_section	A section/sub-section of an Indian Act	Statutes
rule	A rule under a statutory instrument	Statutes
notification	Government notification/gazette entry	Statutes
drug_monograph	Single drug entry from Pharmacopoeia	Pharmacopoeia
formulation	Classical formulation from AFI	Pharmacopoeia
classical_text_verse	Verse/chapter from Charaka/Sushruta/etc.	Classical Texts
patent_record	Patent application/grant record	Patents
trademark_record	Trademark registry entry	IP Registry
gi_record	Geographical Indication entry	IP Registry
case_judgment	Court judgment or order	Case Law
treaty_article	Article from international treaty	Treaties
abs_regulation	ABS rule/regulation/slab	Compliance
regulatory_pathway	Product approval pathway steps	Compliance
tkdl_pointer	Reference to TKDL classification code	TKDL
ingredient_profile	Botanical/mineral substance profile	DRAVYA
food_standard	FSSAI Ayurveda Aahar standard	Regulatory

1.3 Why This Format Works for RAG

```
UDO Record
|
|--> "content" field --> BGE-M3 embeddings --> Qdrant vector store
|
|--> "hierarchy" fields --> Neo4j graph nodes and edges
|
|--> "content" + "title" --> Elasticsearch BM25 index
|
|--> "metadata.citation_format" --> LLM citation enforcement
|
|--> "source" fields --> Audit trail and provenance tracking
```

2. DATA CATEGORY A: Statutes and Rules

2.1 What Exactly Is This Data?

Indian laws (Acts passed by Parliament) and their subordinate rules. These are the PRIMARY knowledge source. The AI must cite these when answering any legal question.

2.2 Exact Statutes to Scrape (17 documents)

#	Statute	Year	Ministry	Why We Need It	Total Sections (approx)
1	Patents Act	1970	Commerce	Core patent law. Sections 3(d), 3(e), 3(p) bar many AYUSH patents	~162 sections
2	Patents Rules	2003 (amended 2024)	Commerce	Procedural rules for filing. RFE timelines, Form 3/27	~195 rules
3	AYUSH Patent Examination Guidelines	2025	AYUSH/IPO	How patent office examines AYUSH applications. Inventive step, synergistic effect, TKDL cross-check	~50 pages
4	Trade Marks Act	1999	Commerce	Brand protection for Ayurvedic product names	~175 sections
5	Geographical Indications Act	1999	Commerce	Region-specific Ayurvedic products (like Darjeeling Tea)	~87 sections
6	Copyright Act	1957	Education	Classical text compilations, Section 2(d) (vi)	~79 sections
7	Designs Act	2000	Commerce	Product packaging protection	~48 sections
8	Plant Variety Protection Act	2001	Agriculture	Cultivar rights, farmers rights	~99 sections
9	Biological Diversity Act	2002 (amended 2023)	Environment	ABS obligations. AYUSH exemptions. Cultivated plant exemptions	~65 sections
10	Biological Diversity Rules	2024	Environment	ABS procedures, digital portal, certificates	~30 rules
11	ABS Regulations	2025	NBA	Benefit-sharing slabs: 0% (<5Cr), 0.2-0.6% (above)	~15 regulations
12	Drugs and Cosmetics Act	1940	Health	Drug/cosmetic regulation. Chapter IVA (ASU drugs). First Schedule texts	~33 sections
13	D&C Rules	1945	Health	Manufacturing, licensing. Rules 151-161 (ASU). Schedule T (GMP)	~200+ rules
14	Drugs and Magic Remedies (OA) Act	1954	Health	Advertising restrictions, prohibited claims	~17 sections

#	Statute	Year	Ministry	Why We Need It	Total Sections (approx)
15	FSSAI Act	2006	Health	Food safety for nutraceuticals/health supplements	~101 sections
16	FSSAI Ayurveda Aahar Regulations	2022	FSSAI	Schedule A traditional food categories	~25 regulations
17	DPDP Act	2023	IT	Data privacy for user interactions	~44 sections

2.3 Exact Data Structure Per Section

Each section of each statute becomes ONE UDO record. Here is exactly what each record looks like:

```

{
  "doc_id": "IND-BDA-2002-S003-SS1",
  "doc_type": "statute_section",
  "title": "Section 3(1) - National Biodiversity Authority",

  "content": "3. National Biodiversity Authority.-(1) With effect from such date as the Central Government may, by notification in the Official Gazette, appoint, there shall be established by the Central Government for the purposes of this Act, a body to be called the National Biodiversity Authority.\n\nProvided that till such time as the National Biodiversity Authority is established, the Central Government shall be competent to take any action or exercise any power under this Act.\n\nExplanation.-For the purposes of this sub-section, \"established\" means established under sub-section (1).",

  "content_plain": "Section 3 Sub-section 1 establishes the National Biodiversity Authority. Until NBA is established the Central Government can exercise powers under this Act.",

  "source": {
    "name": "Biological Diversity Act, 2002",
    "url": "https://indiacode.nic.in/handle/123456789/2046",
    "access_date": "2026-08-15",
    "version": "2023-amendment"
  },

  "hierarchy": {
    "level_1": "Biological Diversity Act, 2002",
    "level_2": "Chapter II - Regulation of Access to Biological Diversity",
    "level_3": "Section 3",
    "level_4": "Sub-section (1)",
    "level_5": null
  },

  "metadata": {
    "jurisdiction": "India",
    "language": "en",
    "category": "statute",
    "sub_category": "biodiversity_law",
    "effective_date": "2004-07-01",
    "last_amended": "2023-08-09",
    "is_current": true,
    "tags": ["NBA", "biodiversity", "ABS", "authority_establishment"],
    "cross_references": ["IND-BDA-2002-S007", "IND-BDA-2002-S038"],
    "provisos": ["Proviso: Central Govt competent until NBA established"],
    "explanations": ["Explanation: 'established' means under sub-section (1)"],
    "keywords_legal": ["shall be established", "Central Government", "notification", "Official Gazette"]
  },

  "rag_config": {
    "chunk_strategy": "section_boundary",
    "max_tokens": 1500,
    "overlap_tokens": 200,
    "embedding_model": "BGE-M3",
    "importance_score": 0.85,
    "citation_format": "[Section 3(1), Biological Diversity Act, 2002]"
  }
}

```

2.4 How to Scrape Statutes

Source: indiacode.nic.in

Method: Step by step

```
Step 1: Navigate to indiacode.nic.in
Step 2: Search by Act name (e.g., "Patents Act, 1970")
Step 3: The site renders HTML with sections in nested div/table structure
Step 4: Use requests + BeautifulSoup to:
  - Extract the Act title, year, ministry
  - For each section:
    - Extract section number (regex: r"^(\d+[A-Z]?)\.")
    - Extract section title (bold text after number)
    - Extract full text including provisos and explanations
    - Detect cross-references (regex: r"section\s+(\d+[A-Z]?)")
    - Detect amendment notes (usually in footnotes or brackets)
Step 5: For each section, create one UDO record
Step 6: Handle special structures:
  - Schedules: Separate UDO per schedule item
  - Provisos: Keep with parent section, tag in metadata
  - Explanations: Keep with parent section, tag in metadata
  - Definitions (usually Section 2): One UDO per defined term
```

Alternate method: Community "Statute API" on GitHub provides pre-structured JSON. Download and transform to UDO format.

Tools needed: requests, beautifulsoup4, re (regex), json

Rate limiting: 1 request per 2 seconds. Total ~17 statutes x ~100 sections = ~1700 records. Takes ~1 hour.

3. DATA CATEGORY B: Pharmacopoeial and Classical Texts

3.1 What Exactly Is This Data?

Official drug standards and classical formulations that define what IS an Ayurvedic medicine. Critical for the Formulation Classifier — the AI needs to know if a product matches a classical text or is proprietary.

3.2 Sub-Category B1: Ayurvedic Pharmacopoeia of India (API) — Drug Monographs

What it looks like in the PDF: Each drug gets 2-5 pages with structured sections. The text is a mix of printed and some handwritten annotations in older volumes. Hindi/Sanskrit names in Devanagari script appear alongside English.

Exact fields to extract per drug monograph:

```

{
  "doc_id": "API-VOL1-ASHWAGANDHA-001",
  "doc_type": "drug_monograph",
  "title": "Ashwagandha (Withania somnifera)",

  "content": "ASHWAGANDHA\nWithania somnifera (Linn.) Dunal.\nFam. Solanaceae\n\nSyns: Sanskrit: Ashwagandha, Varahakarni, Hayagandha\nHindi: Asgandh, Punir\nEnglish: Winter Cherry\n\nPart Used: Root\n\nMacroscopic: Roots are straight, unbranched, varying in length and thickness bearing fibre-like secondary rootlets. Roots are 10-17 cm in length and 6-12 mm in thickness. External surface is buff to grey-yellow with longitudinal wrinkles...\n\nMicroscopic: Transverse section of root shows...\n\nIdentity, Purity and Strength:\nForeign matter: Not more than 2 percent\nTotal Ash: Not more than 7 percent\nAcid-insoluble ash: Not more than 1 percent\nAlcohol-soluble extractive: Not less than 15 percent\nWater-soluble extractive: Not less than 20 percent\n\nChemical Constituents: Withanolides (withaferin A, withanolide D), alkaloids, steroidal lactones\n\nActions: Balya, Vajikara, Rasayana\nTherapeutic Uses: Kshaya, Balakshaya, Vatavyadhi\nDose: 3-6g of the drug in powder form\n\nImportant Formulations: Ashwagandhadi Churna, Ashwagandharishta, Ashwagandha Ghrita",

  "content_structured": {
    "drug_name_sanskrit": "Ashwagandha",
    "drug_name_hindi": "Asgandh",
    "drug_name_english": "Winter Cherry",
    "botanical_name": "Withania somnifera (Linn.) Dunal.",
    "family": "Solanaceae",
    "synonyms": {
      "sanskrit": ["Ashwagandha", "Varahakarni", "Hayagandha"],
      "hindi": ["Asgandh", "Punir"],
      "english": ["Winter Cherry"]
    },
    "part_used": "Root",
    "description_macroscopic": "Roots are straight, unbranched...",
    "description_microscopic": "Transverse section shows...",
    "identity_purity_strength": {
      "foreign_matter": "Not more than 2%",
      "total_ash": "Not more than 7%",
      "acid_insoluble_ash": "Not more than 1%",
      "alcohol_extractive": "Not less than 15%",
      "water_extractive": "Not less than 20%"
    },
    "chemical_constituents": ["Withanolides", "withaferin A", "withanolide D", "alkaloids", "steroidal lactones"],
    "actions_ayurvedic": ["Balya", "Vajikara", "Rasayana"],
    "therapeutic_uses": ["Kshaya", "Balakshaya", "Vatavyadhi"],
    "dose": "3-6g of the drug in powder form",
    "important_formulations": ["Ashwagandhadi Churna", "Ashwagandharishta", "Ashwagandha Ghrita"]
  },

  "source": {
    "name": "Ayurvedic Pharmacopoeia of India, Volume I",
    "url": "https://archive.org/details/ayurvedic-pharmacopoeia-vol1",
    "access_date": "2026-08-15",
    "page_number": "16-18",
    "volume": "Volume I, Part I"
  },

  "hierarchy": {
    "level_1": "Ayurvedic Pharmacopoeia of India",
    "level_2": "Volume I, Part I (Single Drugs)",
    "level_3": "Ashwagandha",
    "level_4": null,

```



```

    "level_5": null
  },

  "metadata": {
    "jurisdiction": "India",
    "language": "en",
    "category": "pharmacopoeia",
    "sub_category": "single_drug_monograph",
    "is_current": true,
    "tags": ["ashwagandha", "withania_somnifera", "solanaceae", "root", "rasayana", "adaptogen"],
    "cross_references": ["AFI-ASHWAGANDHADI-CHURNA", "AFI-ASHWAGANDHARISHTA"],
    "is_in_first_schedule": true,
    "abs_relevance": "Cultivated plant - may be exempt under Biodiversity Amendment 2023",
    "patent_relevance": "Section 3(p) applies if formulation matches classical text"
  },

  "rag_config": {
    "chunk_strategy": "monograph_boundary",
    "max_tokens": 1000,
    "overlap_tokens": 100,
    "embedding_model": "BGE-M3",
    "importance_score": 0.90,
    "citation_format": "[Ashwagandha Monograph, Ayurvedic Pharmacopoeia of India, Vol I]"
  }
}

```

How to scrape this data:

Step 1: Download PDF volumes from archive.org

- Search: "Ayurvedic Pharmacopoeia of India"
- Volumes: Vol I-IX (Part I: Single Drugs), Vol I-III (Part II: Formulations)
- Download as PDF

Step 2: OCR Processing (for scanned volumes)

- Tool: Surya OCR 2 (supports Hindi Devanagari + English mixed text)
- Command: `surya_ocr input.pdf --lang en,hi --output-format markdown`
- This gives you layout-preserving markdown per page

Step 3: Parse OCR output into structured fields

- Each drug monograph starts with the drug name in CAPS
- Regex patterns:
 - Drug name: `r"^[A-Z][A-Z\s]+\n"`
 - Botanical: `r"([A-Z][a-z]+\s[a-z]+\s*\((.*)\)"`
 - Family: `r"Fam\.\s*([A-Za-z]+)"`
 - Synonyms: `r"Syn[s]?:\s*(.*)"(?=Part Used|$)"`
 - Part Used: `r"Part\s+Used:\s*(.*)"`
 - Dose: `r"Dose:\s*(.*)"`
- For each monograph, create one UDO record

Step 4: Quality check

- Verify OCR accuracy on 10% sample
- Manual correction of Sanskrit/Hindi terms
- Cross-check against PCIMH digital versions if available

Volume count: ~600+ drug monographs across all volumes. Each becomes 1 UDO record.

3.3 Sub-Category B2: Ayurvedic Formulary of India (AFI) — Classical Formulations

What it looks like: Each formulation gets 0.5-2 pages listing ingredients, proportions, method, and indications.

Exact fields per formulation:

```

{
  "doc_id": "AFI-PART1-CHURNA-TRIPHALA-001",
  "doc_type": "formulation",
  "title": "Triphala Churna",

  "content": "TRIPHALA CHURNA\nReference: Sarangadhara Samhita, Madhyama Khanda
6/12\n\nIngredients:\n1. Haritaki (Terminalia chebula) - Pericarp - 1 part\n2. Vibhitaki (Terminalia
bellirica) - Pericarp - 1 part\n3. Amalaki (Emblica officinalis) - Pericarp - 1 part\n\nMethod: The
ingredients are cleaned, dried, powdered individually and then mixed together in equal
proportions.\n\nDosage Form: Churna (Powder)\nDose: 3-6g\nAnupana: Warm water, honey, ghee\nTherapeutic
Uses: Constipation (Vibandha), Eye diseases (Netra Roga), Obesity (Sthaulya), Rasayana",

  "content_structured": {
    "formulation_name": "Triphala Churna",
    "formulation_type": "Classical",
    "dosage_form": "Churna (Powder)",
    "reference_text": "Sarangadhara Samhita, Madhyama Khanda 6/12",
    "ingredients": [
      {
        "name_sanskrit": "Haritaki",
        "botanical_name": "Terminalia chebula",
        "part_used": "Pericarp",
        "proportion": "1 part"
      },
      {
        "name_sanskrit": "Vibhitaki",
        "botanical_name": "Terminalia bellirica",
        "part_used": "Pericarp",
        "proportion": "1 part"
      },
      {
        "name_sanskrit": "Amalaki",
        "botanical_name": "Emblica officinalis",
        "part_used": "Pericarp",
        "proportion": "1 part"
      }
    ],
    "method_of_preparation": "Cleaned, dried, powdered individually, mixed in equal proportions",
    "dose": "3-6g",
    "anupana": ["Warm water", "honey", "ghee"],
    "therapeutic_uses": ["Constipation (Vibandha)", "Eye diseases (Netra Roga)", "Obesity (Sthaulya)",
"Rasayana"]
  },

  "source": {
    "name": "Ayurvedic Formulary of India, Part I",
    "url": "https://archive.org/details/afi-part1",
    "page_number": "45",
    "volume": "Part I, Churna Section"
  },

  "metadata": {
    "jurisdiction": "India",
    "language": "en",
    "category": "pharmacopoeia",
    "sub_category": "classical_formulation",
    "is_in_first_schedule": true,
    "classification": "Classical Medicine (Rule 151-161, D&C Rules)",
    "patent_relevance": "Section 3(p) BARS patenting this exact formulation",

```

```

    "regulatory_pathway": "No clinical trial needed for classical formulation"
  },

  "rag_config": {
    "chunk_strategy": "formulation_boundary",
    "max_tokens": 1000,
    "citation_format": "[Triphala Churna, AFI Part I, Ref: Sarangadhara Samhita]"
  }
}

```

Total records: ~1200+ classical formulations across all AFI volumes.

3.4 Sub-Category B3: Classical Texts (e-Samhita)

Source: niimh.nic.in/ebooks (digitized classical texts)

Texts to scrape: Charaka Samhita, Sushruta Samhita, Ashtanga Hridaya, Ashtanga Sangraha, Sarangadhara Samhita, Bhavaprakasha, Madhava Nidana

Format per verse/section:

```

{
  "doc_id": "CS-SUTRA-001-005",
  "doc_type": "classical_text_verse",
  "title": "Charaka Samhita, Sutra Sthana, Chapter 1, Verse 5",
  "content": "Original Sanskrit verse in Devanagari + English translation",
  "content_structured": {
    "text_name": "Charaka Samhita",
    "sthana": "Sutra Sthana",
    "chapter_number": 1,
    "chapter_name": "Deerghanjiviteeya Adhyaya",
    "verse_number": 5,
    "verse_sanskrit": "... (Devanagari text) ...",
    "verse_transliteration": "... (IAST romanization) ...",
    "verse_translation_english": "... (English meaning) ...",
    "commentary": "... (if available) ..."
  },
  "metadata": {
    "category": "classical_text",
    "is_first_schedule_text": true,
    "tags": ["charaka_samhita", "sutra_sthana", "longevity"]
  }
}

```

How to scrape: Playwright (dynamic JS pages) on niimh.nic.in/ebooks. Navigate chapter by chapter, extract verse text and translations.

4. DATA CATEGORY C: Patent and IP Registry Data

4.1 Patent Records (Google BigQuery)

Purpose: Know what AYUSH patents already exist. Helps the IPR Router Agent determine novelty.

BigQuery SQL to get Indian AYUSH patents:

```
SELECT
  pub.publication_number,
  title.text AS title,
  abstract.text AS abstract,
  pub.filing_date,
  pub.publication_date,
  pub.grant_date,
  assignee.name AS assignee,
  ipc.code AS ipc_code,
  cpc.code AS cpc_code,
  pub.family_id
FROM
  `patents-public-data.patents.publications` AS pub,
  UNNEST(pub.title_localized) AS title,
  UNNEST(pub.abstract_localized) AS abstract,
  UNNEST(pub.assignee_harmonized) AS assignee,
  UNNEST(pub.ipc) AS ipc,
  UNNEST(pub.cpc) AS cpc
WHERE
  pub.country_code = 'IN'
  AND title.language = 'en'
  AND abstract.language = 'en'
  AND (
    LOWER(title.text) LIKE '%ayurved%'
    OR LOWER(title.text) LIKE '%herbal%'
    OR LOWER(abstract.text) LIKE '%traditional medicine%'
    OR LOWER(abstract.text) LIKE '%ayush%'
    OR ipc.code LIKE 'A61K36%' -- Medicinal plants
    OR ipc.code LIKE 'A61K35%' -- Animal/plant origin
    OR ipc.code LIKE 'A61P%'   -- Therapeutic activity
  )
LIMIT 50000
```

UDO format for each patent:

```
{
  "doc_id": "PAT-IN-202011001234",
  "doc_type": "patent_record",
  "title": "A novel Ayurvedic composition for management of diabetes",
  "content": "Abstract: The present invention relates to a polyherbal Ayurvedic composition comprising extracts of Gymnema sylvestre, Tinospora cordifolia, and Pterocarpus marsupium in specific ratios...",
  "content_structured": {
    "publication_number": "IN-202011001234-A",
    "filing_date": "2020-01-15",
    "publication_date": "2021-07-23",
    "grant_date": null,
    "status": "Published",
    "assignee": "XYZ Pharmaceuticals Pvt Ltd",
    "inventors": ["Dr. A Kumar", "Dr. B Sharma"],
    "ipc_codes": ["A61K36/48", "A61P3/10"],
    "cpc_codes": ["A61K36/48"],
    "family_id": "72345678",
    "claims_count": 12,
    "key_ingredients": ["Gymnema sylvestre", "Tinospora cordifolia", "Pterocarpus marsupium"]
  },
  "metadata": {
    "category": "patent",
    "sub_category": "ayush_patent",
    "jurisdiction": "India",
    "tags": ["herbal", "diabetes", "polyherbal", "novel_composition"],
    "section_3p_relevant": true,
    "section_3e_relevant": true
  },
  "rag_config": {
    "chunk_strategy": "full_record",
    "citation_format": "[Patent IN-202011001234-A, filed 2020-01-15]"
  }
}
```

4.2 Trademark Records

Source: ipindiaservices.gov.in/tmrpublicsearch (scraped with Playwright)

```
{
  "doc_id": "TM-IN-5678901",
  "doc_type": "trademark_record",
  "content_structured": {
    "application_number": "5678901",
    "trademark_name": "AYURVITAM",
    "class": 5,
    "applicant": "ABC Ayurveda Ltd",
    "filing_date": "2022-03-10",
    "status": "Registered",
    "valid_until": "2032-03-10",
    "goods_services": "Ayurvedic medicines, herbal preparations"
  }
}
```

4.3 Geographical Indication Records

Source: ipindia.gov.in/geographical-indications.htm (PDF lists)

```
{
  "doc_id": "GI-IN-436",
  "doc_type": "gi_record",
  "content_structured": {
    "gi_number": 436,
    "product_name": "Thanjavur Netti",
    "product_type": "Medicinal Product",
    "state": "Tamil Nadu",
    "applicant": "Thanjavur Traditional Healers Association",
    "status": "Registered",
    "class": 5,
    "description": "Traditional Siddha preparation from Thanjavur..."
  }
}
```

5. DATA CATEGORY D: Case Law and Judgments

Source: HuggingFace dataset (d-riti/Dataset-For-Indian-Legal-Knowledge-Base) or IndianKanoon

Filter: Only IPR-related cases — Patents Act, Trademarks Act, GI Act, Biodiversity Act, cases mentioning "Ayurved", "traditional knowledge", "herbal"

```
{
  "doc_id": "CASE-SC-2014-BISWAMOHAN",
  "doc_type": "case_judgment",
  "title": "Biswamohan Mahapatra v. Union of India (2014)",
  "content": "Full text of judgment paragraphs related to traditional knowledge and patent validity...",
  "content_structured": {
    "citation": "AIR 2014 SC 1234",
    "court": "Supreme Court of India",
    "bench": ["Justice X", "Justice Y"],
    "date": "2014-05-15",
    "parties": {
      "petitioner": "Biswamohan Mahapatra",
      "respondent": "Union of India"
    },
    "statutes_cited": ["Patents Act, 1970", "Section 3(p)", "Section 64"],
    "key_issues": ["Whether traditional knowledge can be patented", "Prior art from TKDL"],
    "held": "Patent revoked – formulation found in TKDL as prior art",
    "ratio_decidendi": "A formulation documented in classical texts constitutes prior art under Section 3(p)"
  },
  "metadata": {
    "category": "case_law",
    "sub_category": "patent_ipr",
    "tags": ["section_3p", "TKDL", "prior_art", "patent_revocation"],
    "precedent_value": "High"
  },
  "rag_config": {
    "chunk_strategy": "paragraph_boundary",
    "max_tokens": 2000,
    "overlap_tokens": 300,
    "citation_format": "[Biswamohan v. UOI, AIR 2014 SC 1234]"
  }
}
```

6. DATA CATEGORY E: Treaties and International Law

Source: WIPO Lex (wipo.int/wipolex), CBD ABSCH API

Exact treaties to scrape (8 documents):

Treaty	Articles	Key for AYUSH
TRIPS Agreement	~73 articles	Art 27 (patentable), Art 29 (disclosure)
Convention on Biological Diversity	~42 articles	Art 8(j) (TK), Art 15 (access)
Nagoya Protocol	~36 articles	PIC, MAT, checkpoints
WIPO GRATK Treaty 2024	~20 articles	Mandatory disclosure of country-of-origin
PCT (Patent Cooperation Treaty)	~69 articles	International patent filing
Madrid Protocol	~18 articles	International trademark

Treaty	Articles	Key for AYUSH
EU THMPD	~25 articles	EU herbal medicines pathway
US DSHEA 1994	~15 sections	US dietary supplement rules

UDO format per treaty article:

```
{
  "doc_id": "TREATY-TRIPS-ART027",
  "doc_type": "treaty_article",
  "title": "Article 27 - Patentable Subject Matter (TRIPS)",
  "content": "1. Subject to the provisions of paragraphs 2 and 3, patents shall be available for any inventions, whether products or processes, in all fields of technology, provided that they are new, involve an inventive step and are capable of industrial application...",
  "content_structured": {
    "treaty_name": "TRIPS Agreement",
    "article_number": 27,
    "article_title": "Patentable Subject Matter",
    "paragraphs": [
      {"number": 1, "text": "...patents shall be available..."},
      {"number": 2, "text": "Members may exclude from patentability..."},
      {"number": 3, "text": "Members may also exclude from patentability..."}
    ],
    "india_position": "India implements via Patents Act 1970, with broader exclusions under Section 3"
  },
  "metadata": {
    "jurisdiction": "International",
    "category": "treaty",
    "signatories_count": 164,
    "india_ratified": true,
    "tags": ["patentability", "inventive_step", "industrial_application"]
  },
  "rag_config": {
    "citation_format": "[Article 27, TRIPS Agreement (WTO)]"
  }
}
```

7. DATA CATEGORY F: Regulatory and Compliance Data

7.1 ABS Slabs (Biological Diversity Regulations)

```
{
  "doc_id": "ABS-SLAB-2025-001",
  "doc_type": "abs_regulation",
  "title": "ABS Benefit Sharing Rate - Turnover below Rs 5 Crore",
  "content": "Under the ABS Regulations 2025, companies with annual turnover below Rs 5 Crore using wild biological resources are required to intimate the NBA but the benefit sharing rate is 0% (nil).",
  "content_structured": {
    "user_type": "Company",
    "turnover_slab": "Below Rs 5 Crore",
    "resource_type": "Wild biological resources",
    "abs_rate": "0%",
    "filing_requirement": "Intimation to NBA",
    "portal": "absefiling.nbaindia.in",
    "exemptions": [
      "Registered AYUSH practitioners using codified TK - FULLY EXEMPT",
      "Cultivated medicinal plants - EXEMPT (Certificate of Origin needed)"
    ]
  },
  "metadata": {
    "category": "compliance",
    "sub_category": "abs_regulation",
    "tags": ["ABS", "benefit_sharing", "biodiversity", "turnover_slab"]
  },
  "rag_config": {
    "citation_format": "[ABS Regulations, 2025 - Turnover Slab <5Cr]"
  }
}
```

7.2 DRAVYA Ingredient Profiles

Source: ccras.nic.in (Playwright scraping)

```
{
  "doc_id": "DRAVYA-WITHANIA-001",
  "doc_type": "ingredient_profile",
  "title": "Ashwagandha - Ingredient Profile",
  "content_structured": {
    "name_sanskrit": "Ashwagandha",
    "name_hindi": "Asgandh",
    "name_english": "Indian Ginseng / Winter Cherry",
    "botanical_name": "Withania somnifera",
    "family": "Solanaceae",
    "part_used": ["Root", "Leaf"],
    "rasa": "Tikta (Bitter), Kashaya (Astringent)",
    "guna": "Laghu (Light), Snigdha (Unctuous)",
    "viryā": "Ushna (Hot)",
    "vipaka": "Madhura (Sweet)",
    "dosha_action": "Vata-Kapha shamaka",
    "therapeutic_uses": ["Rasayana", "Balya", "Vajikara", "Nidrajanana"],
    "habitat": "Drier regions of India, Madhya Pradesh, Rajasthan",
    "cultivation_status": "Widely cultivated",
    "abs_status": "Cultivated - likely exempt under BDA 2023 Amendment",
    "schedule_status": "Listed in First Schedule texts",
    "tkdl_code_hint": "TKRC A/pharmaceutical/plant-origin/solanaceae"
  }
}
```

8. DATA CATEGORY G: TKDL Reference Data

CANNOT directly scrape TKDL (restricted). But we CAN build a pointer system.

```
{
  "doc_id": "TKDL-POINTER-ASHWAGANDHA",
  "doc_type": "tkdl_pointer",
  "title": "TKDL Reference - Ashwagandha Formulations",
  "content": "The Traditional Knowledge Digital Library contains documented formulations using Ashwagandha from classical Ayurvedic texts. These serve as prior art for patent examination. To search: visit tkdl.res.in, use TKRC classification code A (Ayurveda), navigate to pharmaceutical preparations containing Withania somnifera.",
  "content_structured": {
    "ingredient": "Ashwagandha (Withania somnifera)",
    "tkrc_section": "A (Ayurveda)",
    "tkrc_class": "Pharmaceutical preparations (Kalpana)",
    "estimated_formulations": "50+",
    "search_guidance": "Navigate TKRC: A > Pharmaceutical > Plant-origin > Solanaceae > Withania somnifera",
    "ipc_mapping": "A61K36/81 (Solanaceae)",
    "prior_art_impact": "Any formulation matching TKDL entry cannot be patented under Section 3(p)",
    "access_instructions": "TKDL access requires subscription. Contact tkdl.res.in for institutional access."
  },
  "metadata": {
    "category": "tkdl_reference",
    "tags": ["TKDL", "prior_art", "ashwagandha", "section_3p"]
  }
}
```

9. Scraping Methodology — Step by Step Per Source

9.1 Source-by-Source Scraping Playbook

Source	Tool	Authentication	Rate Limit	Pages/Records	Time Estimate	Output
India Code	requests + BS4	None	1 req/2s	~1700 sections	~1 hour	1700 UDOs
Gazette PDFs	Direct download + PyMuPDF	None	N/A	~50 PDFs	~30 min	~500 UDOs
AYUSH Guidelines PDF	Direct download + PyMuPDF	None	N/A	1 PDF (~50 pages)	~10 min	~30 UDOs
WIPO Lex	Partial API + PDF download	None	N/A	~8 treaties	~1 hour	~300 UDOs
CBD ABSCH	REST API	API key (free)	100 req/min	~500 records	~10 min	~500 UDOs
API/AFI PDFs	archive.org download + Surya OCR	None	N/A	~12 volumes, ~3000 pages	~6 hours OCR	~1800 UDOs
e-Samhita	Playwright	None	1 req/3s	~5000 verses	~4 hours	~5000 UDOs
DRAVYA Portal	Playwright	None	1 req/3s	~500 entries	~30 min	~500 UDOs
FSSAI Aahar	PDF download + PyMuPDF	None	N/A	1 regulation	~10 min	~50 UDOs
Google Patents BigQuery	BigQuery SQL	Google Cloud (free tier)	1TB/month free	~50K patents	~5 min query	~50K UDOs
WIPO PATENTSCOPE	REST API	API key (free)	100 req/min	~5K PCT apps	~1 hour	~5K UDOs
InPASS	Playwright	None	1 req/5s (careful)	~2K patents	~3 hours	~2K UDOs
TMR India	Playwright	None	1 req/5s	~1K marks	~1.5 hours	~1K UDOs
GI Registry	PDF download	None	N/A	~10 PDF lists	~15 min	~100 UDOs
IndianKanoon/HF	HuggingFace datasets library	None	N/A	~20K judgments (filtered)	~10 min download	~20K UDOs

Source	Tool	Authentication	Rate Limit	Pages/Records	Time Estimate	Output
NBA ABS Portal	Playwright	None	1 req/5s	~200 filings	~20 min	~200 UDOs
KanoonGPT HF	HuggingFace datasets	None	N/A	Bulk download	~5 min	Supplement
Ayusoft	Playwright	None	1 req/3s	~300 entries	~15 min	~300 UDOs

9.2 Total Data Volume

Metric	Value
Total UDO Records	~89,000
Total Raw Data	~5-8 GB (PDFs, HTML, JSON)
Total Processed Data	~3 GB (UDO JSON + embeddings)
Vector Store (Qdrant)	~860 MB (89K x 1024-dim vectors)
BM25 Index (Elasticsearch)	~500 MB
Knowledge Graph (Neo4j)	~2 GB (100K nodes, 500K edges)

10. Complete Field Reference

10.1 All UDO Fields — Master Dictionary

Field	Type	Required	Description
<code>doc_id</code>	string	YES	Unique ID: SOURCE-TYPE-IDENTIFIER
<code>doc_type</code>	enum	YES	One of 16 document types (see Section 1.2)
<code>title</code>	string	YES	Human-readable title for display
<code>content</code>	string	YES	Full text content (for embedding and BM25)
<code>content_plain</code>	string	NO	Stripped/simplified version
<code>content_structured</code>	object	RECOMMENDED	Parsed fields specific to doc_type
<code>source.name</code>	string	YES	Name of source document/database
<code>source.url</code>	string	YES	URL where data was obtained
<code>source.access_date</code>	date	YES	When data was scraped
<code>source.version</code>	string	NO	Amendment/edition version

Field	Type	Required	Description
<code>source.page_number</code>	string	NO	Page in original document
<code>source.volume</code>	string	NO	Volume/part number
<code>hierarchy.level_1</code>	string	YES	Top-level grouping (Act name, Book name)
<code>hierarchy.level_2</code>	string	NO	Chapter/Part
<code>hierarchy.level_3</code>	string	NO	Section/Article
<code>hierarchy.level_4</code>	string	NO	Sub-section/Clause
<code>hierarchy.level_5</code>	string	NO	Proviso/Explanation
<code>metadata.jurisdiction</code>	enum	YES	"India" or "International" or specific country
<code>metadata.language</code>	string	YES	ISO 639-1 code (en, hi, ta)
<code>metadata.category</code>	enum	YES	statute, pharmacopoeia, patent, case_law, treaty, compliance, tkdl_reference
<code>metadata.sub_category</code>	string	YES	Specific type within category
<code>metadata.effective_date</code>	date	NO	When provision took effect
<code>metadata.last_amended</code>	date	NO	Last amendment date
<code>metadata.is_current</code>	boolean	YES	Whether this is the current version
<code>metadata.tags</code>	array[string]	YES	Searchable tags and keywords
<code>metadata.cross_references</code>	array[string]	NO	doc_ids of related records
<code>rag_config.chunk_strategy</code>	string	YES	How to chunk this record
<code>rag_config.max_tokens</code>	int	YES	Maximum chunk size
<code>rag_config.overlap_tokens</code>	int	YES	Overlap for continuity
<code>rag_config.importance_score</code>	float	NO	0-1, how critical this is (for ranking boost)
<code>rag_config.citation_format</code>	string	YES	Exact citation string for LLM to use
<code>checksum</code>	string	NO	SHA-256 hash for deduplication and change detection

10.2 Data Flow: Raw Source to RAG-Ready

```
RAW SOURCE (PDF/HTML/JSON/API)
  |
  v
COLLECTOR (Playwright/BS4/BigQuery/HF)
  |
  v
PROCESSOR (PyMuPDF/Surya OCR/HTML Parser)
  |
  v
PARSER (Regex + Custom Section Splitter)
  |
  v
UDO BUILDER (Python script creates JSON records)
  |
  v
VALIDATOR (JSON Schema validation, dedup check)
  |
  v
GIT COMMIT (corpus/ directory, version tracked)
  |
  v
EMBEDDING PIPELINE
  |--- content field --> BGE-M3 --> Qdrant (vector search)
  |--- content + title --> Elasticsearch (BM25 keyword search)
  |--- hierarchy + cross_references --> Neo4j (graph relations)
  |--- rag_config.citation_format --> stored for LLM prompt injection
```

Document Version: 1.0

Date: August 26, 2026

Total Sources: 23 data sources

Total Records: ~89,000 UDO records

Unified Format: Universal Document Object (UDO) JSON